BIO 203 BIO TUTORIAL

Associations Between Reef Disturbance, Habitat Characteristics, and Sea Urchin Species Richness Across Selected Philippine Marine Sites

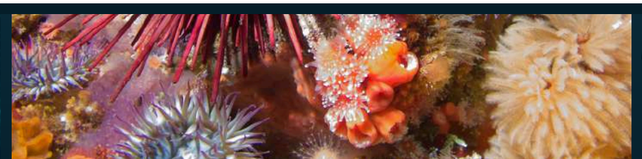
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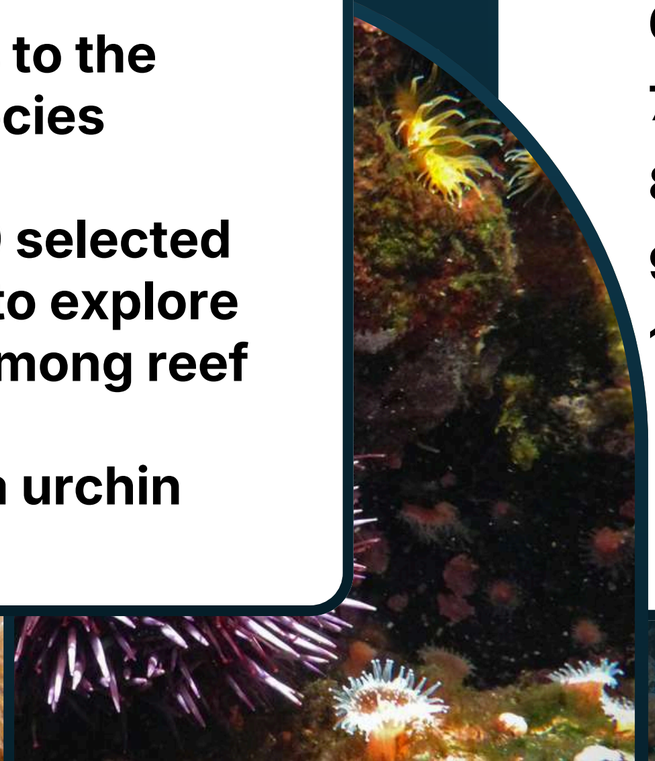
INTRODUCTION

- Coral reefs support diverse marine organisms and provide important habitats.
- Reef disturbance and habitat characteristics may influence marine biodiversity.
- Sea urchins are marine invertebrates that form part of reef communities.
- Species richness refers to the number of different species recorded in an area.
- This study compares 10 selected Philippine marine sites to explore possible associations among reef disturbance, habitat characteristics, and sea urchin species richness.

10 Marine Sites

1. Apo Island, Negros Oriental
2. Sumilon Island, Cebu
3. San Jose Marine Reserve, Negros Oriental
4. Pamilacan Island, Bohol
5. Centro Marine Sanctuary, Bohol
6. Popabion Marine Sanctuary, Liloan, Cebu
7. Cantagay Marine Sanctuary, Bohol
8. Malapascua Island, northern Cebu
9. Sillión Marine Sanctuary, Bantayan Island
10. Danjugan Island Sanctuary

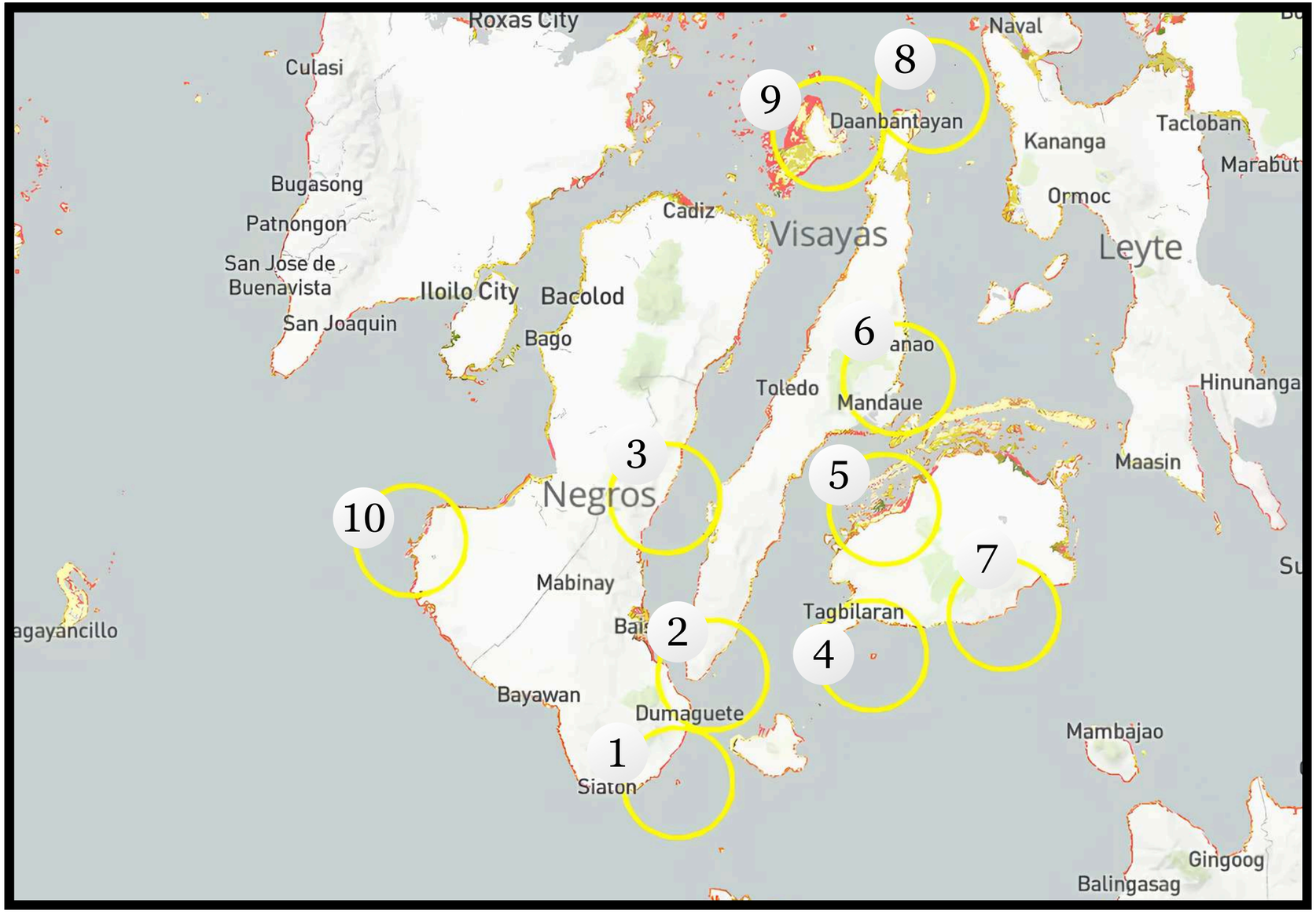






MAP OF THE CHOSEN 10 SITES





STEP 1

Select and document the 10 sites

site_id	site_name	latitude	longitude	disturbance_rank
1	Apo Island	9.07605	123.27015	2
2	Sumilon Island	9.430983	123.386267	6
3	San Jose Marine Reserve	10.012778	123.226889	8
4	Pamilacan Island	9.495833	123.917222	7
5	Centro Marine Sanctuary	9.97584	123.95817	4
6	Poblacion Marine Sanctuary, Liloan	10.404323	124.004518	9
7	Cantagay Marine Sanctuary	9.630219	124.358211	3
8	Malapascua Island	11.332222	124.118889	10
9	Sillion Marine Sanctuary	11.20901	123.77075	5
10	Danjugan Island Sanctuary	9.8729	122.3789	1

STEP 1

	Evidence	Rationale
2/10	Apo Island - Russ GR, Lowe JR, Rizzari JR, Bergseth BJ, Alcala AC. Partitioning no-take marine reserve (NTMR) and benthic habitat effects on density of small and large-bodied tropical wrasses. PLoS One. 2017 Dec 7;12(12):e0188515. doi: 10.1371/journal.pone.0188515. PMID: 29216194; PMCID: PMC5720769.	Apo was assigned a low rank because it has a long history of marine-reserve protection and community support. Long-term research documented increases in predatory reef-fish density and biomass during sustained protection. However, this does not mean Apo is free from tourism, fishing outside protected zones, or other pressures.
6/10	Sumilon Island - A. C. Alcala, G. R. Russ, A direct test of the effects of protective management on abundance and yield of tropical marine resources, Journal du Conseil, Volume 47, Issue 1, 1990, Pages 40–47, https://doi.org/10.1093/icesjms/47.1.40	Sumilon was assigned a moderate rank because its management history includes periods of effective protection followed by periods when fishing resumed. Research documented declines in reef-fish abundance and fishery yield after protection broke down. This provides evidence that disturbance can increase when management is not maintained, although it does not independently establish the site's current rank
8/10	San Jose Marine Reserve - Coral Triangle Initiative. (n.d.). San Jose Marine Reserve. Coral Triangle Atlas. https://ctatlas.coraltriangleinitiative.org/MPA?id=4022200	San Jose was assigned a high provisional rank in the initial working classification. However, the available sanctuary record does not provide sufficient direct evidence to establish that its disturbance is high. The rank should be treated as uncertain until site-specific information on coral condition, fishing, tourism, water quality, and enforcement can be obtained
7/10	Pamilacan Island - White, A., Molina, D., Delizo, D., Jr., Diaz, R., & Sabonsolin, A. (2024). Summary field report: “Saving Philippine Reefs” coral reef monitoring expedition to Bohol. Coastal Conservation and Education Foundation, Inc. https://www.coast.ph/publications/summary-field-report-saving-philippine-reefs-coral-reef-monitoring-expedition-to-bohol/	Pamilacan was assigned a moderately high rank because historical monitoring documented a decrease in live hard-coral cover between 1984 and 1999, including a decline inside the sanctuary. However, more recent monitoring reports improved coral cover and ongoing local management. Therefore, the evidence indicates a history of ecological pressure, not proof of high current disturbance.
4/10	Centro Marine Sanctuary - Bohol, Philippines A.C. Alcala, A. Bucol and P. Nillos (2008).Directory of Marine Reserves in the Visayas, Philippines.Silliman University Press, Dumaguete City.	Centro was assigned a relatively low-to-moderate rank based mainly on its recorded marine-sanctuary designation. However, designation alone does not demonstrate effective enforcement, healthy coral, or low human pressure. Without comparable ecological monitoring, this rank is a working estimate rather than a confirmed result.

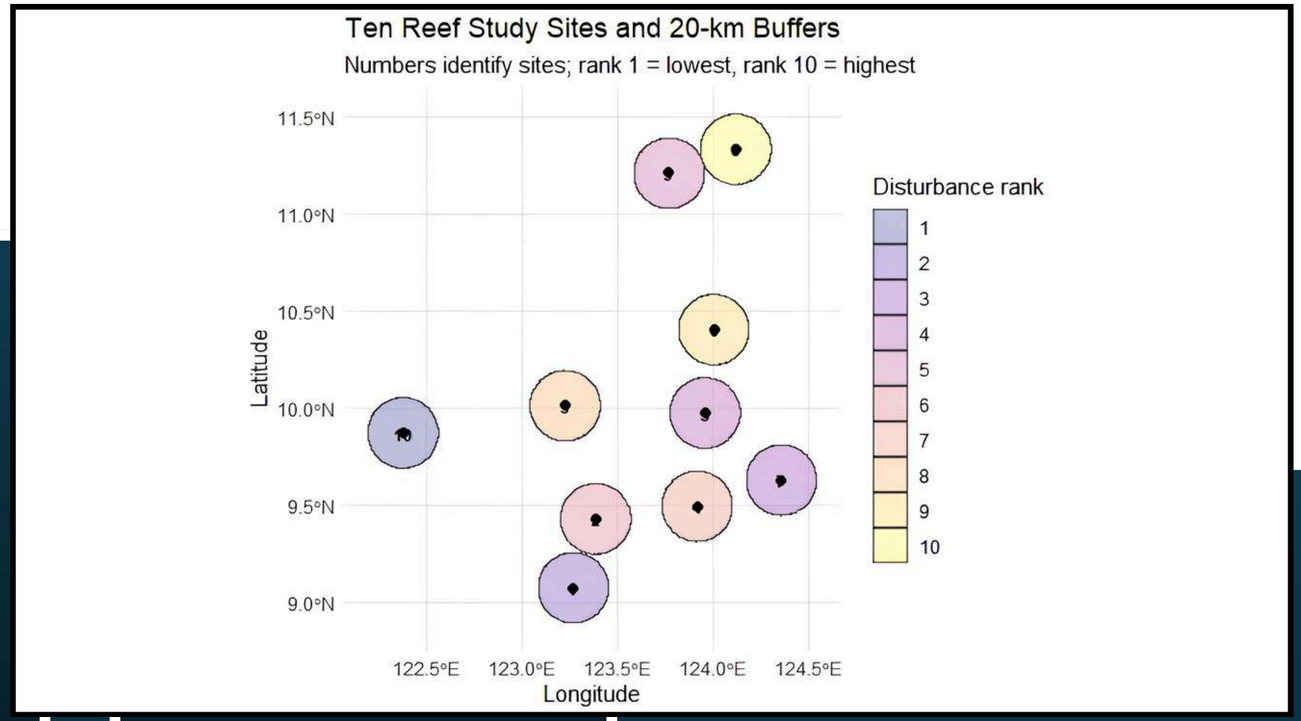
STEP 1

	Evidence	Rationale
9/10	Poblacion Marine Sanctuary, Liloan - Edullantes B, Maglangit F, Ortiz AM, Casibo JMR, Vicentuan LLC, Bensig EO (2021) A benchmark survey of plankton, fish and benthic composition in Poblacion and Kadurong Reefs in Liloan, Cebu, Philippines. Biodiversity Data Journal 9: e72537. https://doi.org/10.3897/BDJ.9.e72537	Poblacion was assigned a high rank in the initial classification, but the available reef-monitoring study is not sufficient by itself to establish high disturbance. Monitoring activity is not evidence of degradation, and the rank requires stronger, directly comparable data on ecological condition and human pressures.
3/10	Cantagay Marine Sanctuary - Alcala, A. C., Bucol, A. A., & Nillos-Kleiven, P. (2008). Directory of marine reserves in the Visayas, Philippines. Foundation for the Philippine Environment and Silliman University-Angelo King Center for Research and Environmental Management. https://www.researchgate.net/publication/279860997_Directory_of_Marine_Reserves_in_the_Visayas_Philippines	Cantagay was assigned a relatively low rank because historical sanctuary records describe a functioning marine sanctuary and provide baseline information on coral cover and fish biomass. These records offer some indication of reef condition and management, but they are dated and do not prove that present-day disturbance is low.
10/10	Malapascua Island - People and the Sea. (2019). Assessment of Crown-of-Thorns (COTs) seastar population around the island of Malapascua, Daanbantayan, Philippines. https://www.peopleandthesea.org/wp-content/uploads/2019/10/PepSea_ReefMonitoringeReport2018.pdf	Malapascua was assigned the highest rank because published local monitoring has reported reef damage, concerns about overharvesting, and pressures associated with human activities. Separate assessments have also documented substantial coral damage following typhoons. These sources support the presence of multiple pressures, but they do not prove Malapascua is more disturbed than every other site in the study.
5/10	Sillion Marine Sanctuary - Alcala, A. C., Bucol, A. A., & Nillos-Kleiven, P. (2008). Directory of marine reserves in the Visayas, Philippines. Foundation for the Philippine Environment and Silliman University-Angelo King Center for Research and Environmental Management. https://www.researchgate.net/publication/279860997_Directory_of_Marine_Reserves_in_the_Visayas_Philippines	Sillion was placed in the middle of the ranking because historical records report fair coral cover and local management arrangements. These suggest that the site has both conservation measures and potential ecological pressures. However, the available older data cannot establish its current disturbance relative to the other sites.
1/10	Danjugan Island Sanctuary - Wright, L.S. (2022). Small no-take areas benefit hard corals more than regulation through fishing permits. Aquatic Conservation: Marine and Freshwater Ecosystems, 32(7), 1209–1216. https://doi.org/10.1002/aqc.3814	Danjugan was ranked lowest because it has a documented conservation program and no-take marine reserves. Research found substantially greater hard-coral abundance inside no-take zones than in areas where fishing was regulated through permits. This supports the value of stronger protection, although it does not establish that the entire surrounding 20-km buffer has the lowest disturbance.

STEP 2

Create the same 20-km buffer for Allen Coral Atlas and GBIF





STEP 3 Allen Coral Atlas: describe reef topography (geomorphic zonation)

SITE ID	DESCRIPTION
1	Reef slope was the largest mapped geomorphic category (52.7%), followed by reef flat (41.0%).
2	Reef slope accounted for 48.3% and reef flat for 40.6% of the mapped area. Lagoon made up 8.4%.
3	Reef flat dominated at 69.1%, while reef slope accounted for 19.0% and lagoon 10.3%.
4	Reef flat represented 61.3% of the mapped area, followed by reef slope at 26.5% and lagoon at 9.3%.
5	Reef slope accounted for 39.7% and reef flat 34.1%. Other geomorphic features made up 17.4%, while lagoon represented 8.7%.
6	Reef flat was strongly dominant at 74.9%, followed by reef slope at 16.1% and lagoon at 6.8%.
7	Reef slope represented 47.5% and reef flat 44.2%, with lagoon at 6.9%.
8	Reef flat accounted for 43.4%, reef slope 29.1%, and other geomorphic features 20.0%. Lagoon represented 7.3%.
9	Reef flat represented 44.8% and reef slope 34.5%. Other geomorphic features accounted for 16.1%, while lagoon represented 4.4%.
10	Reef flat accounted for 48.4% and reef slope 32.2%. Other geomorphic features represented 16.6%, while lagoon made up 2.5%.

STEP 4 Allen Coral Atlas: quantify benthic composition

Site	Largest benthic class	Percentage
Apo Island	Coral/Algae	70.08%
Sumilon Island	Coral/Algae	52.51%
San Jose Marine Reserve	Coral/Algae	31.73%
Pamilacan Island	Coral/Algae	34.89%
Centro Marine Sanctuary	Coral/Algae	48.92%
Poblacion Marine Sanctuary, Liloan	Rubble	32.43%
Cantagay Marine Sanctuary	Coral/Algae	48.25%
Malapascua Island	Coral/Algae	39.57%
Sillion Marine Sanctuary	Coral/Algae	48.19%
Danjugan Island Sanctuary	Coral/Algae	64.32%

STEP 5 Human population density and turbidity

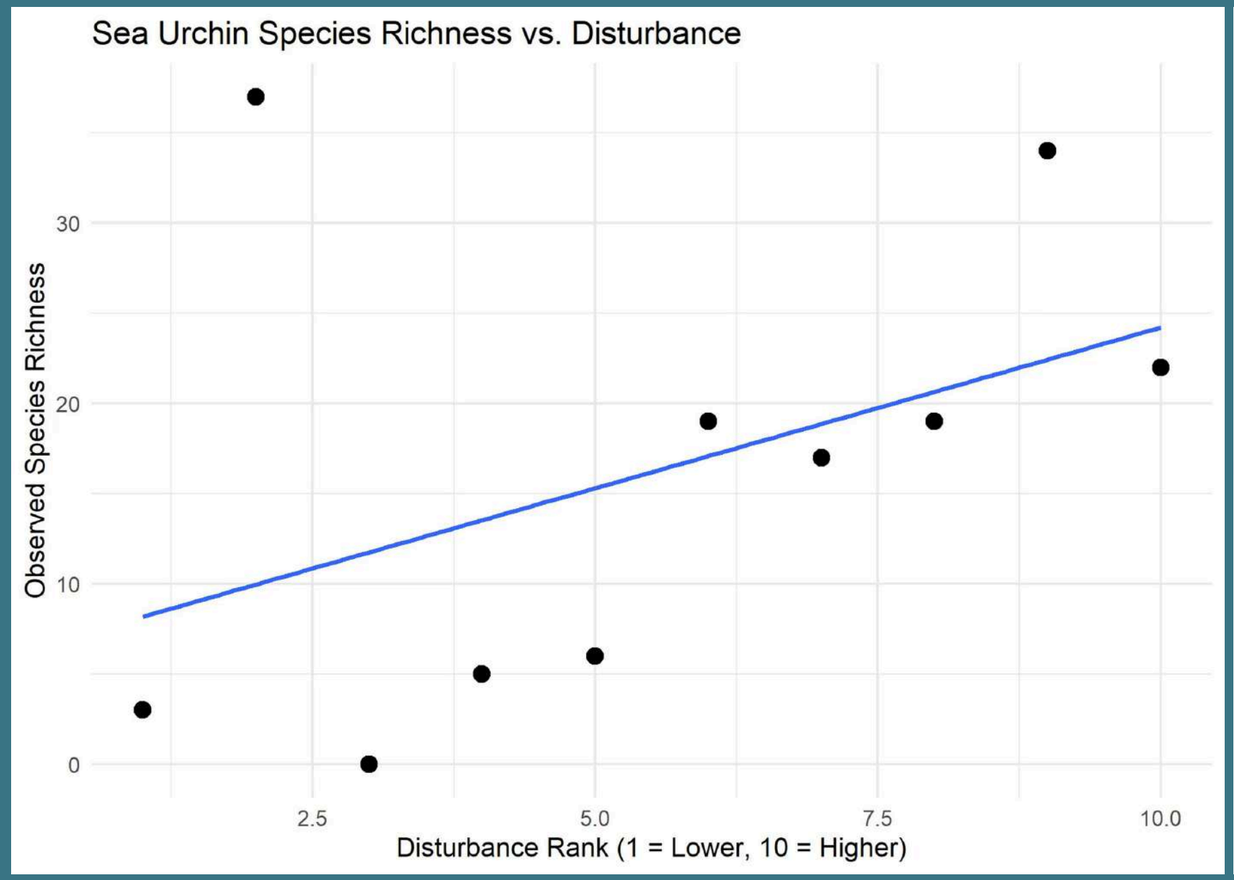
Site	Reef site	Mean population density (people/km²)	Median turbidity (FNU)
1	Apo Island	453.82	5.9
2	Sumilon Island	1,052.47	5.7
3	San Jose Marine Reserve	443.55	5.5
4	Pamilacan Island	1,132.41	4.5
5	Centro Marine Sanctuary	543.57	5.1
6	Poblacion Marine Sanctuary, Liloan	4,595.28	4.8
7	Cantagay Marine Sanctuary	360.95	5.4
8	Malapascua Island	1,164.43	5.9
9	Sillion Marine Sanctuary	1,690.45	5.9
10	Danjugan Island Sanctuary	296.18	6.4

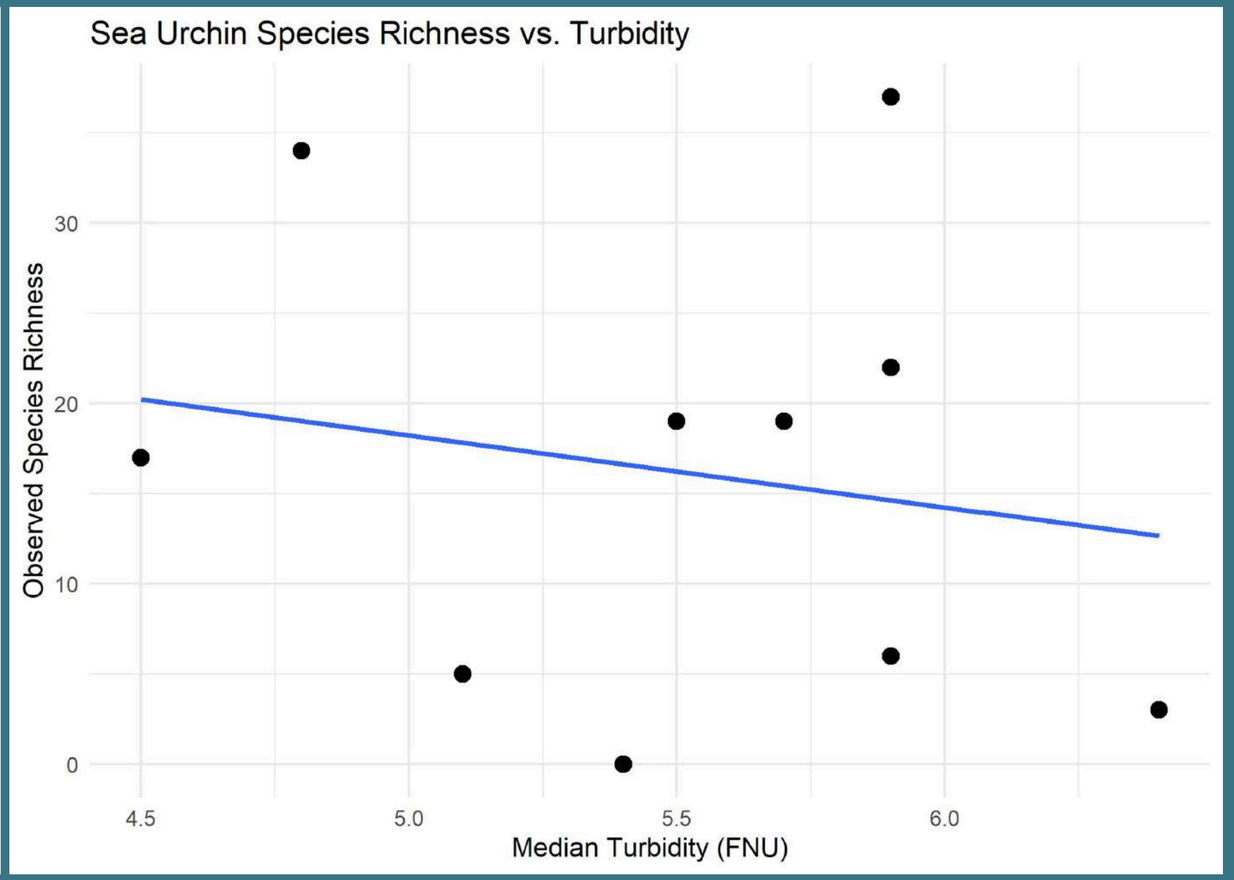
STEP 6 GBIF multi-species extraction in R

Site	Site name	Retained records	Observed species richness
1	Apo Island	214	37
2	Sumilon Island	57	19
3	San Jose Marine Reserve	74	19
4	Pamilacan Island	88	17
5	Centro Marine Sanctuary	7	5
6	Poblacion Marine Sanctuary, Liloan	88	34
7	Cantagay Marine Sanctuary	0	0
8	Malapascua Island	286	22
9	Sillion Marine Sanctuary	10	6
10	Danjugan Island Sanctuary	6	3

STEP 7

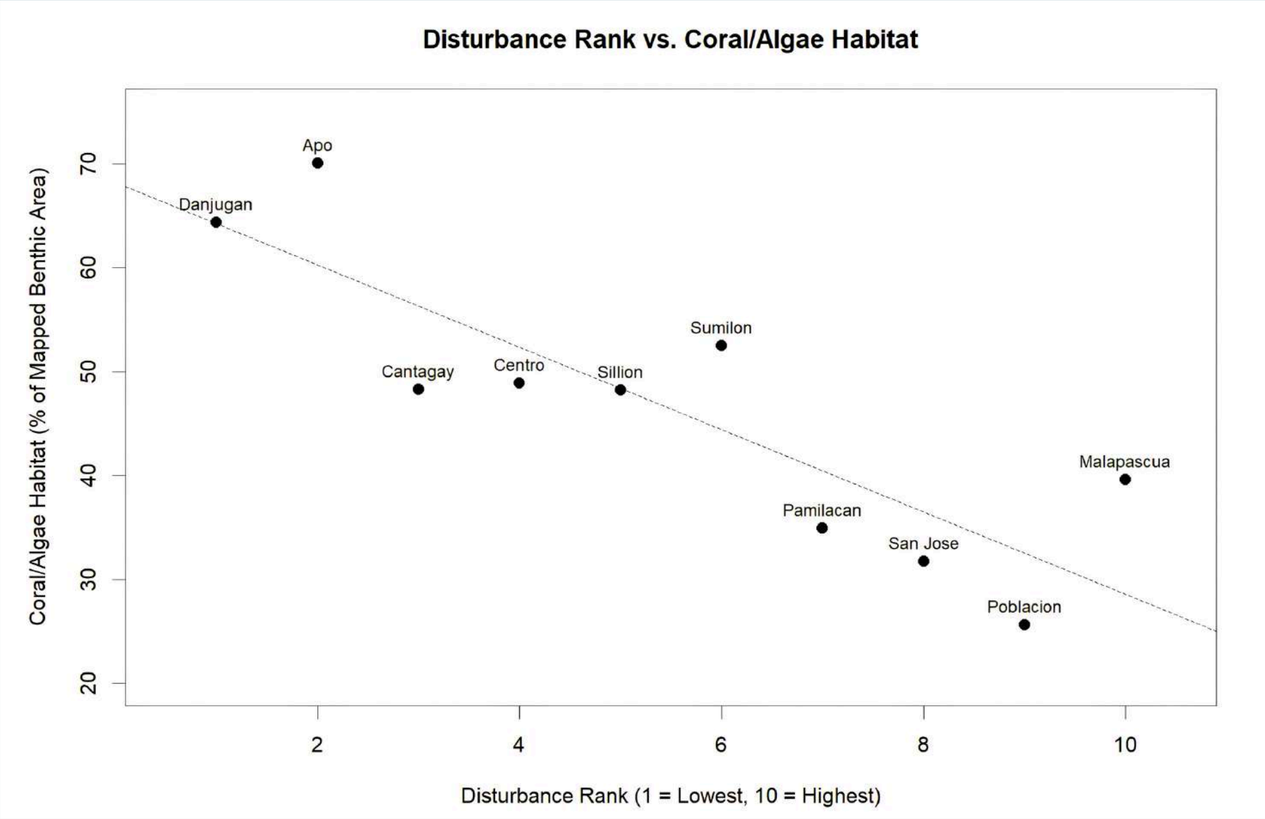
Relationships Between Urchins Species Richness and Environmental Factors

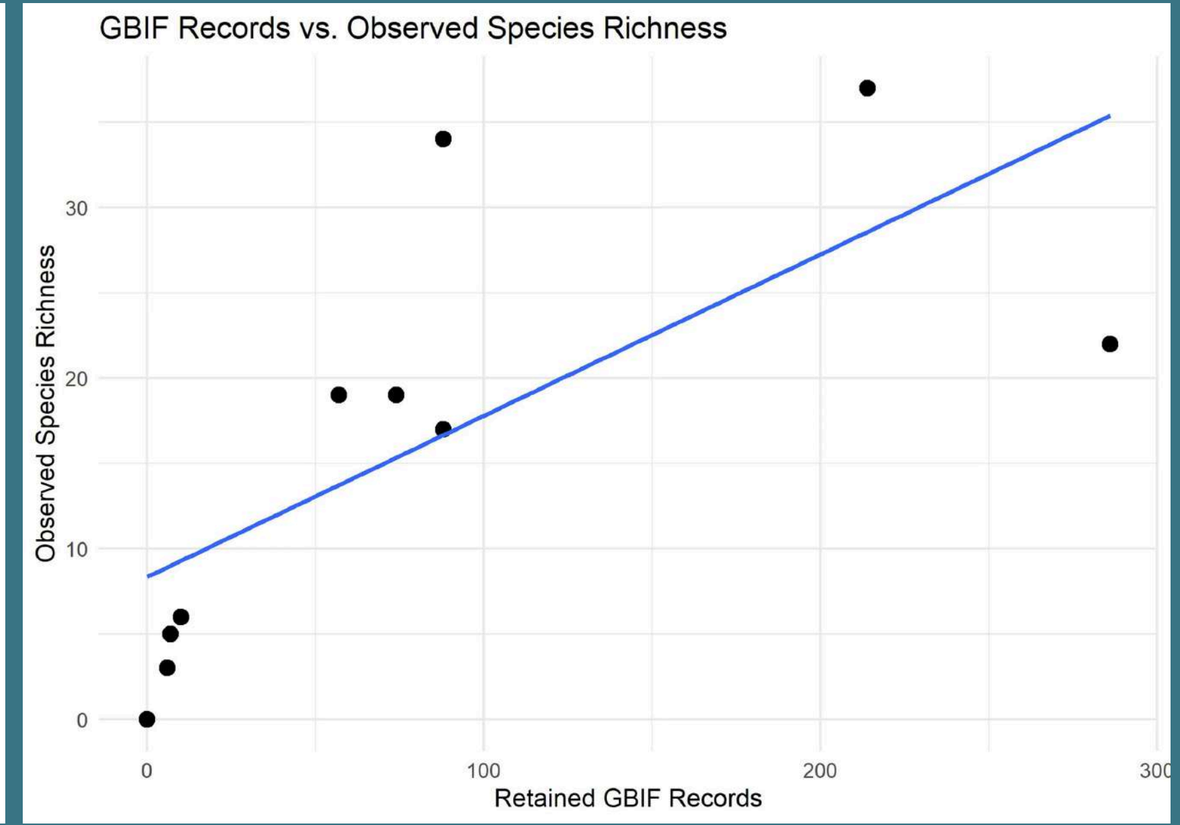




STEP 7

Relationships Between Urchins Species Richness and Environmental Factors





STEP 7

SPEARMAN CORRELATION SUMMARY

Variables compared	Spearman's ρ	p-value	Interpretation
Richness vs. Coral/Algae %	−0.201	0.5784	Very weak negative association
Richness vs. Disturbance rank	0.505	0.1369	Moderate positive association
Richness vs. Median turbidity	0.012	0.9731	Almost no association

FINAL MULTIVARIABLE MODEL

Predictor	Estimate	p-value	Interpretation
Coral/Algae percentage	−0.044	0.980	No clear evidence of an association
Median turbidity	−0.218	0.869	No clear evidence of an association

STEP 7

Poisson Regression: Disturbance Rank vs. Observed Species Richness

Model statistic	Result
Predictor	Disturbance rank
Coefficient estimate	0.11230
p-value	0.0000691
Null deviance	104.96
Residual deviance	88.62
Residual degrees of freedom	8
AIC	132.77
Overdispersion ratio	11.08

Coefficient Results

Predictor	Estimate	Interpretation
Intercept	2.1159	Expected baseline on the model's log scale
Disturbance rank	0.1123	Positive association with observed species richness

Reef topography /geomorphic patterns



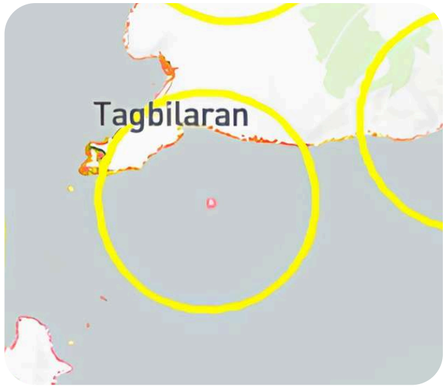
1. Apo Island, Negros Oriental



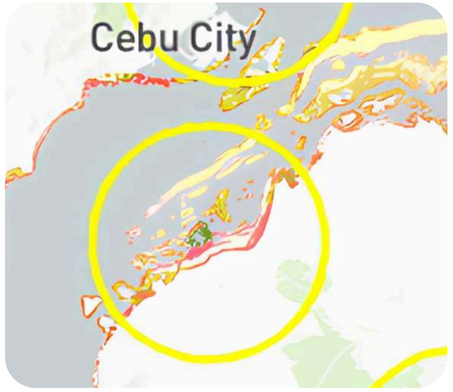
2. Sumilon Island, Cebu



3. San Jose Marine Reserve, Negros Oriental



4. Pamilacan Island, Bohol 5.



Centro Marine Sanctuary, Bohol



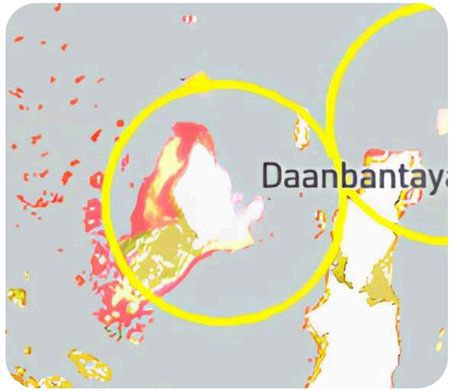
6.Poblacion Marine Sanctuary, Liloan, Cebu



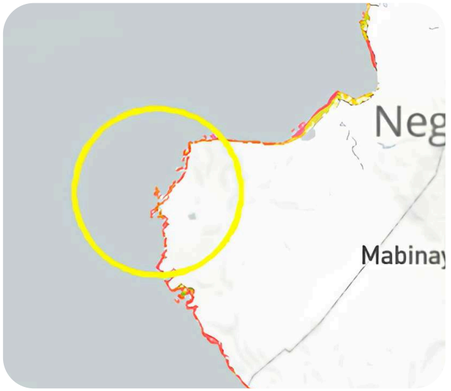
7. Cantagay Marine Sanctuary, Bohol



8. Malapascua Island, northern Cebu



9. Sillión Marine Sanctuary, Bantayan



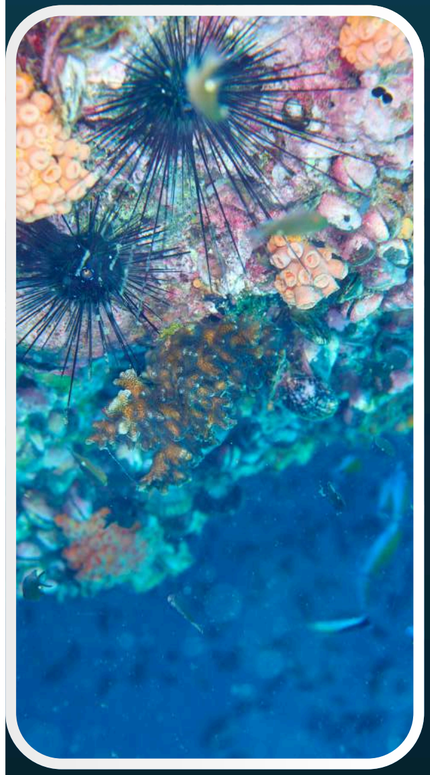
10. Danjugan Island Sanctuary

Interpretation

- Reef disturbance levels differ among the 10 selected Philippine marine sites.
- Marine protected areas and no-take zones can help protect coral reefs and support marine biodiversity.
- Historical reef damage and human activities may contribute to differences in reef conditions among sites.
- Reef geomorphology varies across sites, with reef flats and reef slopes being the major habitat features.
- Differences in habitat characteristics may influence the distribution and diversity of marine organisms, including sea urchins.
- Community participation, environmental education, and sustainable tourism can contribute to reef conservation.
- The disturbance rankings are preliminary and should not be interpreted as confirmed measurements of current reef conditions.

QUESTIONS

- 1. What ecological disturbance gradient did your 10 sites represent, and what evidence supports that gradient?
- 2. How did reef topography difer among low- and high-disturbance sites?
- 3. Which benthic classes dominated each end of the gradient?
- 4. How did turbidity and nearby human population density vary among sites?
- 5. How many GBIF records and species were obtained at each site?
- 6. Were sites with high species richness also sites with high GBIF sampling efort?
- 7. Which environmental variable showed the clearest relationship with species richness for your taxonomic group?
- 8. Did your simple GLM/regression support the group’s ecological hypothesis? Describe direction and uncertainty, not just the p-value. 9. What is one plausible biological explanation for the pattern?
- 10. What are at least three reasons the pattern might be biased or confounded?



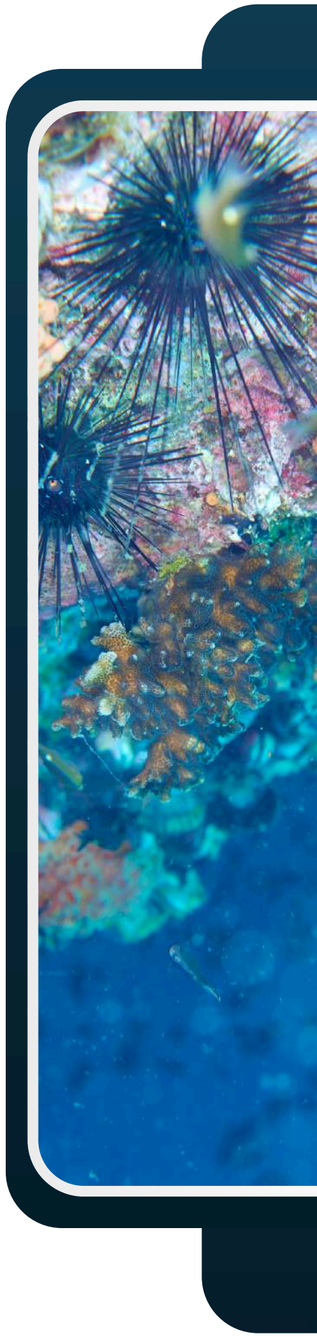
1. The 10 Philippine marine sites were assigned preliminary disturbance rankings from 1/10 to 10/10. Danjugan Island Sanctuary ranked lowest (1/10) because of its conservation programs and no-take zones, followed by Apo Island (2/10), which has a long history of marine protection. Cantagay, Centro, Sillión, Sumilon, and Pamilacan received rankings of 3/10, 4/10, 5/10, 6/10, and 7/10, respectively, based on historical sanctuary records, management history, and documented coral-cover changes. San Jose and Poblacion received rankings of 8/10 and 9/10, although evidence supporting their high disturbance levels was insufficient. Malapascua ranked highest (10/10) because of documented reef damage and pressures from human activities. However, these rankings are preliminary because they rely mainly on historical studies rather than direct measurements of current reef conditions.

2. The Allen Coral Atlas data showed that reef flats and reef slopes were the dominant geomorphic features, but their proportions varied among sites. Danjugan Island Sanctuary had 48.4% reef flats and 32.2% reef slopes, while Apo Island had 52.7% reef slopes and 41.0% reef flats. Malapascua Island had 43.4% reef flats, 29.1% reef slopes, and 20.0% other geomorphic features. Poblacion Marine Sanctuary had 74.9% reef flats, 16.1% reef slopes, and 6.8% lagoons. These results show that reef topography varied across the sites, but geomorphic differences alone cannot establish the level of ecological disturbance.

3. The presentation includes a section on benthic composition using Allen Coral Atlas data, but the actual percentages and dominant benthic classes are not clearly provided in the available results. Therefore, I cannot accurately identify which benthic classes dominated the lowest- and highest-ranked sites. The original benthic composition data are needed to make this comparison without assuming or inventing results.

4. The presentation identifies turbidity and nearby human population density as environmental variables considered in the comparison of the 10 sites. However, the corresponding slide does not provide readable numerical values or a clear site-by-site comparison. Therefore, I cannot determine which sites had the highest or lowest turbidity or population density. Although these factors may be associated with reef conditions, the available data are insufficient to establish a relationship between them and the assigned disturbance rankings.

5. The presentation states that GBIF data were extracted using R, but the exact number of records and species for each of the 10 sites is not visible in the provided results. Therefore, I cannot accurately report the site-specific counts without the original extraction table or R output. These numbers are important because differences in GBIF records may reflect sampling effort and data availability rather than actual differences in species richness.



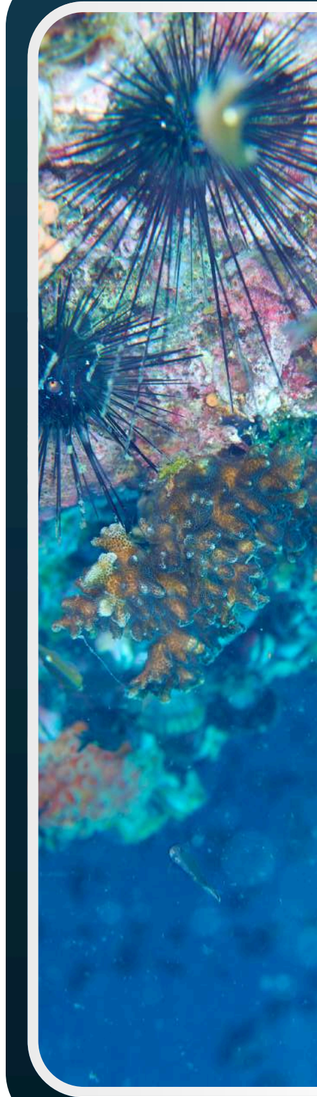
6. Yes, sites with higher GBIF sampling effort generally had higher recorded species richness. Malapascua Island had the highest number of retained records, with 286 records and 22 species, while Apo Island had 214 records and 37 species.

7. Disturbance rank showed the clearest statistical relationship with sea urchin species richness in our analysis. The results indicated a positive association, meaning that sites with higher disturbance ranks tended to have higher observed species richness. However, other environmental factors, such as reef topography, benthic composition, turbidity, and human population density, may also influence the distribution of sea urchin species.

8. Yes, the Poisson regression supported our hypothesis that reef disturbance is associated with sea urchin species richness. The regression coefficient was positive at 0.1123, indicating that higher disturbance ranks were associated with higher observed species richness. However, the overdispersion ratio of 11.08 suggests substantial variation in the data that the model does not adequately explain. Therefore, although the positive relationship was statistically significant, the result should be interpreted cautiously because other factors may also influence species richness.

9. One plausible biological explanation is the intermediate disturbance hypothesis, which suggests that moderate levels of disturbance may allow more species to coexist by preventing some species from dominating the community. Different sea urchin species may also have different environmental tolerances, allowing them to occupy habitats with varying disturbance levels. However, further investigation is needed to determine whether this hypothesis explains the observed pattern in our study sites.

10. The observed pattern may be affected by several factors. First, uneven sampling effort means that some sites have substantially more GBIF records than others, potentially increasing their recorded species richness. Second, the absence of GBIF records at Cantagay Marine Sanctuary does not necessarily mean that sea urchins are absent, since the site may simply lack sufficient sampling. Third, confounding environmental variables, such as wave exposure, water depth, substrate type, and food availability, may independently influence sea urchin species richness. Additionally, differences in data quality and the use of historical disturbance rankings may affect the reliability of comparisons among sites.



Limitations

- Disturbance rankings are based mainly on existing studies and historical records rather than direct field measurements.
- Some references contain outdated information that may not represent present-day reef conditions.
- Available ecological data are not equally detailed or comparable across all 10 sites.
- Some sites lack sufficient evidence on coral health, fishing pressure, tourism, water quality, and enforcement.
- Habitat descriptions alone cannot establish the actual diversity or abundance of sea urchins.
- The presentation does not show sufficient direct sea urchin survey data to confirm relationships between disturbance, habitat characteristics, and species richness.
- The results should be interpreted as preliminary patterns rather than proof of cause-and-effect relationships.





Conclusion

- Reef conditions and habitat characteristics vary among the 10 selected Philippine marine sites.
- Marine protection and effective management are important for conserving coral reef ecosystems.
- Human activities and environmental disturbances may affect reef habitats and the organisms that depend on them.
- Further field surveys are needed to determine sea urchin species richness and validate the proposed relationships.
- Consistent monitoring of reef conditions and habitat characteristics can improve comparisons among sites.
- Community participation, responsible tourism, and environmental education are essential for long-term reef conservation.
- Protecting coral reefs helps maintain marine biodiversity and ecosystem health.



Thankyou for paying attention

Thank You & Protect Our Oceans Together

